AMS 526: Numerical Linear Algebra for Computational and Data Sciences

Lecture 2: Algorithmic Considerations; Orthogonality

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Outline

1. **Algorithmic Considerations (MC §1.2)**
2. Orthogonal Vectors and Matrices (NLA §2)

Algorithms for Matrix-Vector Multiplication

- Suppose $A \in \mathbb{R}^{m \times n}$ and $x \in \mathbb{R}^n$.
- MATLAB-style code for $b = Ax$:

Row oriented

```
for i = 1 : m
  b(i) = 0
  for j = 1 : n
    b(i) = b(i) + A(i, j) * x(j)
  end
end
```

Column oriented

```
b(:) = 0
for j = 1 : n
  for i = 1 : m
    b(i) = b(i) + A(i, j) * x(j)
  end
end
```

- Number of operations is $\mathcal{O}(mn)$, but big-O notation is insufficient.
- Number of operations is approximately $2mn$: the coefficient of the leading-order term is important for comparison.

Flop Count

- It is important to assess the *efficiency* of algorithms. But how?
  - We could implement different algorithms and perform direct comparisons, but implementation details can affect true performance.
  - We could estimate the cost of all operations, but it is very tedious.
  - A relatively simple and effective approach is to estimate the amount of floating-point operations, or "flops", and focus on asymptotic analysis as the sizes of matrices approach infinity.
- Idealization:
  - Count each operation $+$, $-$, $*$, $/$, and $\sqrt{\ }$ as one flop.
  - This estimation is crude, as it omits data movement in memory, which is non-negligible on modern computer architectures (e.g., different loop orders can affect cache performance).
- Matrix-vector product requires about $2mn$ flops.
- Suppose $m = n$; it takes quadratic time in n, or $\mathcal{O}\left(n^2\right)$.

Algorithms for Saxpy: Scalar a x plus y

- Saxpy computes $ax + y$ and updates y:

$$y = ax + y \implies y_i = ax_i + y_i.$$

- Suppose $x, y \in \mathbb{R}^n$ and $a \in \mathbb{R}$.

MATLAB-style code

```
for i = 1 : n
    y(i) = y(i) + a * x(i)
end
```

Pseudo-code

```
for i = 1 : n
    y_i ← y_i + a x_i
```

- Number of flops is $2n$.
- Pseudo-code cannot run on any computer, but is human-readable and straightforward to convert into actual code in any programming language (e.g., C, Fortran, MATLAB, Python).
- We use pseudo-code on slides for conciseness.

Gaxpy: Generalized Saxpy

- Computes $y = y + Ax$, where $A \in \mathbb{R}^{m \times n}$ and $x \in \mathbb{R}^n$.

Row oriented

for $i = 1 : m$
  for $j = 1 : n$
    $y_i = y_i + a_{ij} x_j$

Column oriented

for $j = 1 : n$
  for $i = 1 : m$
    $y_i = y_i + a_{ij} x_j$

- Inner loop of column-oriented algorithm can be converted to vector operation $y = y + x_j a_{:,j}$.
- Number of flops is $2mn$.

Matrix Multiplication Update

- Computes $C = C + AB$, where $A \in \mathbb{R}^{m\times r}$, $B \in \mathbb{R}^{r\times n}$, and $C \in \mathbb{R}^{m\times n}$:

$$c_{ij} = c_{ij} + \sum_{k=1}^{r} a_{ik} b_{kj}.$$

Standard Update Loop

```
for i = 1 : m
  for j = 1 : n
    for k = 1 : r
      c_ij = c_ij + a_ik b_kj
```

- The operation further generalizes Gaxpy.
- Number of flops is $2mnr$.
- In BLAS (Basic Linear Algebra Subroutines), functions are grouped into Level-1, 2, and 3, depending on whether complexity is linear, quadratic, or cubic.

Six Variants of Algorithms

- There are six variants depending on permutation of i, j, and k:

$$ijk, \quad jik, \quad ikj, \quad jki, \quad kij, \quad kji.$$

- Inner product version: $c_{ij} = c_{ij} + a_{i,:}b_{:,j}$ (or $a_{i,:}b_j$).

Inner Product Version

for $i = 1 : m$
  for $j = 1 : n$
    $c_{ij} = c_{ij} + a_{i,:}b_j$

- Saxpy version: computes as $c_{:,j} = c_{:,j} + Ab_{:,j}$.

Saxpy Version

for $j = 1 : n$
  $c_{:,j} = c_{:,j} + Ab_{:,j}$

Six Variants of Algorithms (Cont.)

- Outer product version: computes as $C = C + \sum_{k=1}^{r} a_{:,k} b_{k,:}$.

Outer Product Version

for $k = 1 : r$
  $C = C + a_{:,k} b_{k,:}$

Outline

1. Algorithmic Considerations (MC §1.2)
2. **Orthogonal Vectors and Matrices (NLA §2)**

Inner Product

- *Euclidean length* of $\boldsymbol{u}$ is the square root of the inner product of $\boldsymbol{u}$ with itself, i.e., $\|\boldsymbol{u}\|_2 = \sqrt{\boldsymbol{u}^\top \boldsymbol{u}}$.
- The inner product of two **unit** vectors $\boldsymbol{u}$ and $\boldsymbol{v}$ is the cosine of the angle α between $\boldsymbol{u}$ and $\boldsymbol{v}$, i.e., $\cos\alpha = \boldsymbol{u}^\top \boldsymbol{v}$.
- The inner product is *bilinear*, in the sense that it is linear in each vector separately:

$$
\begin{aligned}
(u_1 + u_2)^\top v &= u_1^\top v + u_2^\top v, \\
u^\top (v_1 + v_2) &= u^\top v_1 + u^\top v_2, \\
(\alpha u)^\top (\beta v) &= \alpha\beta u^\top v.
\end{aligned}
$$

Orthogonal Vectors

Definition

Vectors x and y are *orthogonal* if $x^\top y = 0$.

In other words, the angle between them is $90°$.

Definition

Two sets of vectors X and Y are orthogonal if every $x \in X$ is orthogonal to every $y \in Y$.

- Subspace $S^\perp$ is the *orthogonal complement* of S if they are orthogonal and complementary subspaces.
- For $A \in \mathbb{R}^{m \times n}$, $\text{null}(A)$ is the *orthogonal complement* of $\text{range}(A^\top)$.

Definition

A set of nonzero vectors S is *orthogonal* if they are pairwise orthogonal. They are *orthonormal* if the set is orthogonal and, in addition, each vector has unit Euclidean length.

Orthogonal Vectors and Independence

Theorem

The vectors in an orthogonal set S are linearly independent.

Proof

Prove by contradiction. If a vector can be expressed as a linear combination of the other vectors in the set, then it is orthogonal to itself, which implies it is the zero vector (contradicting the definition of S).

Question: If the column vectors of an $m \times n$ matrix A are orthogonal, what is the rank of A?

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Question: If the column vectors of an $m \times n$ matrix A are orthogonal, what is the rank of A?
Answer: n (assuming $m \geq n$). Since the columns are orthogonal, they are linearly independent. Thus, A has full rank.

Components of Vector

- Given an orthonormal set $\{q_1, q_2, \dots, q_m\}$ forming a basis of $\mathbb{R}^m$, any vector v can be decomposed into orthogonal components as $v = \sum_{i=1}^{m} (q_i^\top v) q_i$.
- Another way to express the decomposition is $v = \sum_{i=1}^{m} (q_i q_i^\top) v$.
- $q_i q_i^\top$ is an *orthogonal projection matrix*.
- More generally, given an orthonormal set $\{q_1, q_2, \dots, q_n\}$ with $n \leq m$, we have

$$v = r + \sum_{i=1}^{n} (q_i^\top v) q_i = r + \sum_{i=1}^{n} (q_i q_i^\top) v,$$

  where $r^\top q_i = 0$ for $1 \leq i \leq n$.
- Let Q be composed of column vectors $\{q_1, q_2, \dots, q_n\}$. Then $P = QQ^\top = \sum_{i=1}^{n} (q_i q_i^\top)$ is an orthogonal projection matrix (more in Lecture 4).
- **Question**: What is $Q^\top Q$ equal to?

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- **Question**: What is $Q^\top Q$ equal to?
- **Answer**: $Q^\top Q = I_n$.

Orthogonal Matrices

Definition

A square matrix Q is *orthogonal* if $Q^\top = Q^{-1}$, i.e., if $Q^\top Q = QQ^\top = I$.

- Its column vectors are *orthonormal*. In other words, $q_i^\top q_j = \delta_{ij}$, the *Kronecker delta*.
- For complex matrices, we say the matrix is *unitary* if $Q^H = Q^{-1}$.

Question: What is the geometric meaning of multiplication by an orthogonal matrix?

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Question: What is the geometric meaning of multiplication by an orthogonal matrix?

Answer: It preserves angles and Euclidean lengths. In the real case, multiplication by an orthogonal matrix Q corresponds to a rotation (if $\det(Q) = 1$) or a reflection (if $\det(Q) = -1$).